

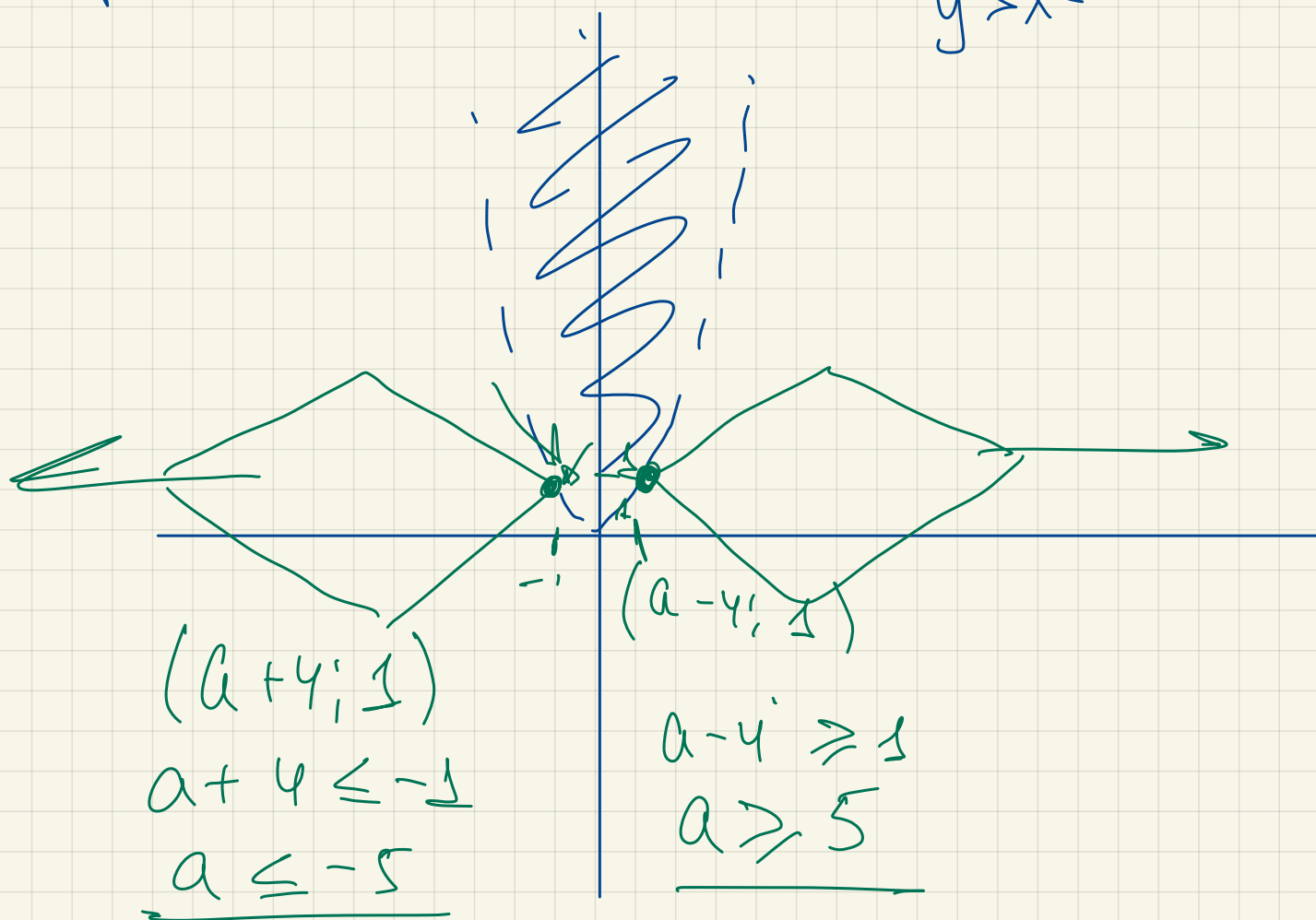


$$\begin{cases} (|x-a| + 2|y-1| - 4)(x^2 + (y-4)^2 - 1) \ln(y-x^2) = 0 \\ (|x-a| + 2|y-1| - 4) \left(\frac{|a|}{15} (x-4) - y + 5 \right) = 0 \end{cases}$$

4 pen

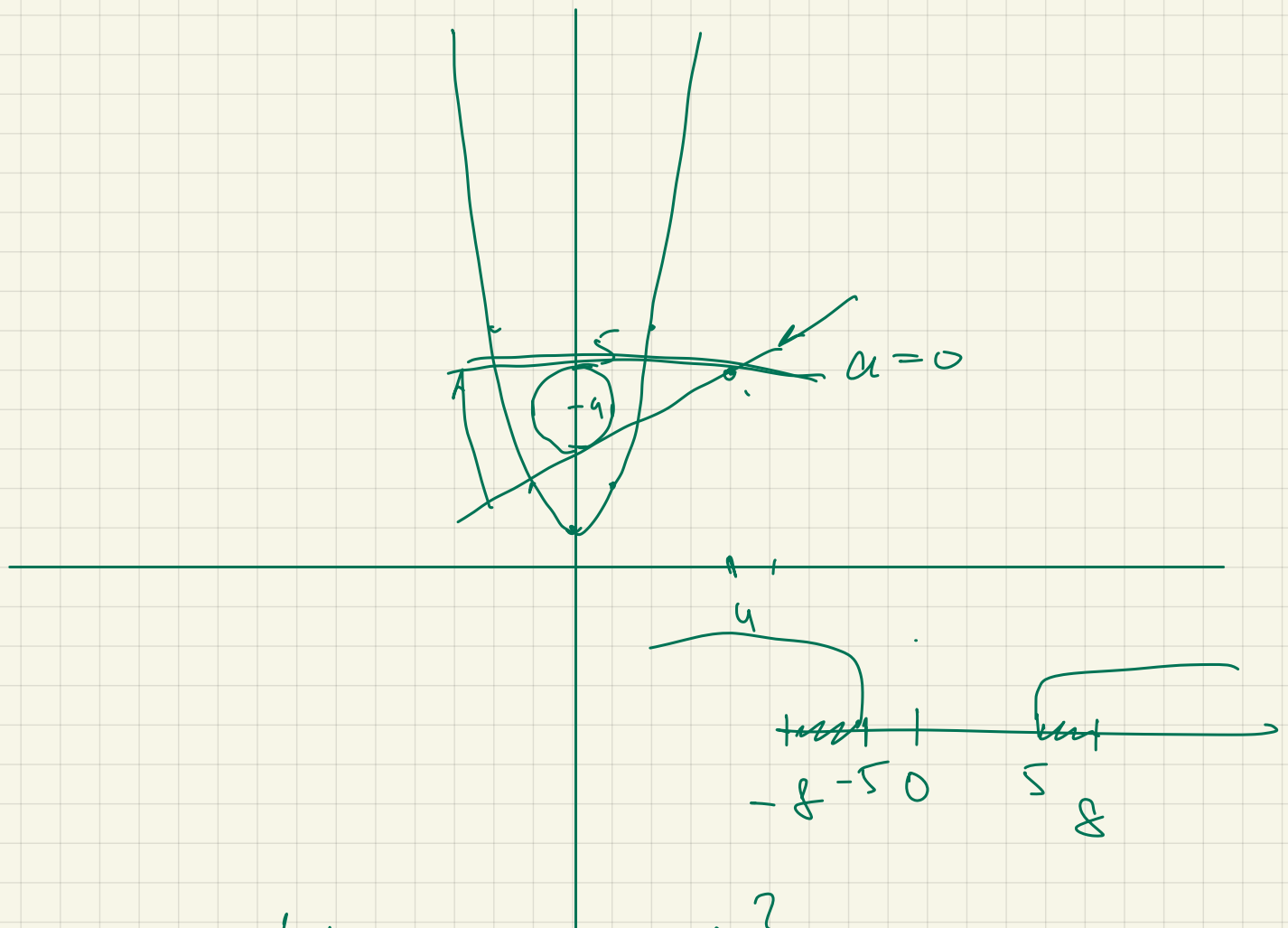
$$y - x^2 > 0$$

$$y > x^2$$



$$\begin{cases} (x^2 + (y-4)^2 - 1) \ln(y-x^2) = 0 \\ \frac{|a|}{15} (x-4) - y + 5 = 0 \end{cases}$$

$$\begin{cases} x^2 + (y-4)^2 = 1 \\ y = x^2 + 1 \\ \frac{|a|}{15} |x-4| - y + 5 = 0 \\ a \in (-\infty; -5) \cup (5; +\infty) \end{cases} \quad y = \frac{|a|}{15} |x-4| + 5 \quad (4; 5)$$



$$x^2 + \left(\frac{|a|}{15} |x-4| + 5 - 4 \right)^2 = 1$$

1 peu ?

$$x^2 + \left(\frac{|a|}{15} (x-4) \right)^2 + \cancel{4} + \frac{2|a|}{15} (x-4) = \cancel{4}$$

$$x^2 + \frac{a^2}{15^2}(x^2 - 8x + 16) + \frac{2|a|}{15}(x-4) = 0$$

$$x^2 \left(1 + \frac{a^2}{15^2} \right) - x \left(\frac{8a^2}{15^2} - \frac{2|a|}{15} \right) + \frac{a^2 16}{15^2} - \frac{8|a|}{15} = 0$$

$$\left(\frac{8a^2}{15^2} - \frac{2|a|}{15} \right)^2 - 4 \cdot \left(1 + \frac{a^2}{15^2} \right) \left(\frac{a^2 16}{15^2} - \frac{8|a|}{15} \right) = 0$$

$$\frac{64a^4}{15^4} - 2 \cdot \frac{8a^2}{15^2} \frac{2|a|}{15} + \frac{4a^2}{15^2} - 4 \left(\frac{a^2 16}{15^2} - \frac{8|a|}{15} \right) + \frac{a^4 \cdot 16}{15^4} - \frac{8|a| \cdot a^2}{15^3} = 0$$

$$\cancel{\frac{64a^4}{15^4}} - \cancel{\frac{32}{15^3} a^2 |a|} + \frac{4a^2}{15^2} - \frac{64a^2}{15^2} + \frac{32|a|}{15} - \cancel{\frac{64a^4}{15^4}} + \cancel{\frac{32|a|a^2}{15^3}} = 0$$

$$\frac{32|a|}{15} - \frac{60a^2}{15^2} = 0$$

$$8|a| - \frac{15a^2}{15} = 0$$

$$8|a| - a^2 = 0$$

$$a^2 - 8|a| = 0$$

$$a \geq 0$$

$$a^2 - 8a = 0$$

$$a = 0$$

$$a = 8$$

$$a \leq 0$$

$$a^2 + 8a = 0$$

$$a = 0$$

$$a = -8$$

$$a = 0$$

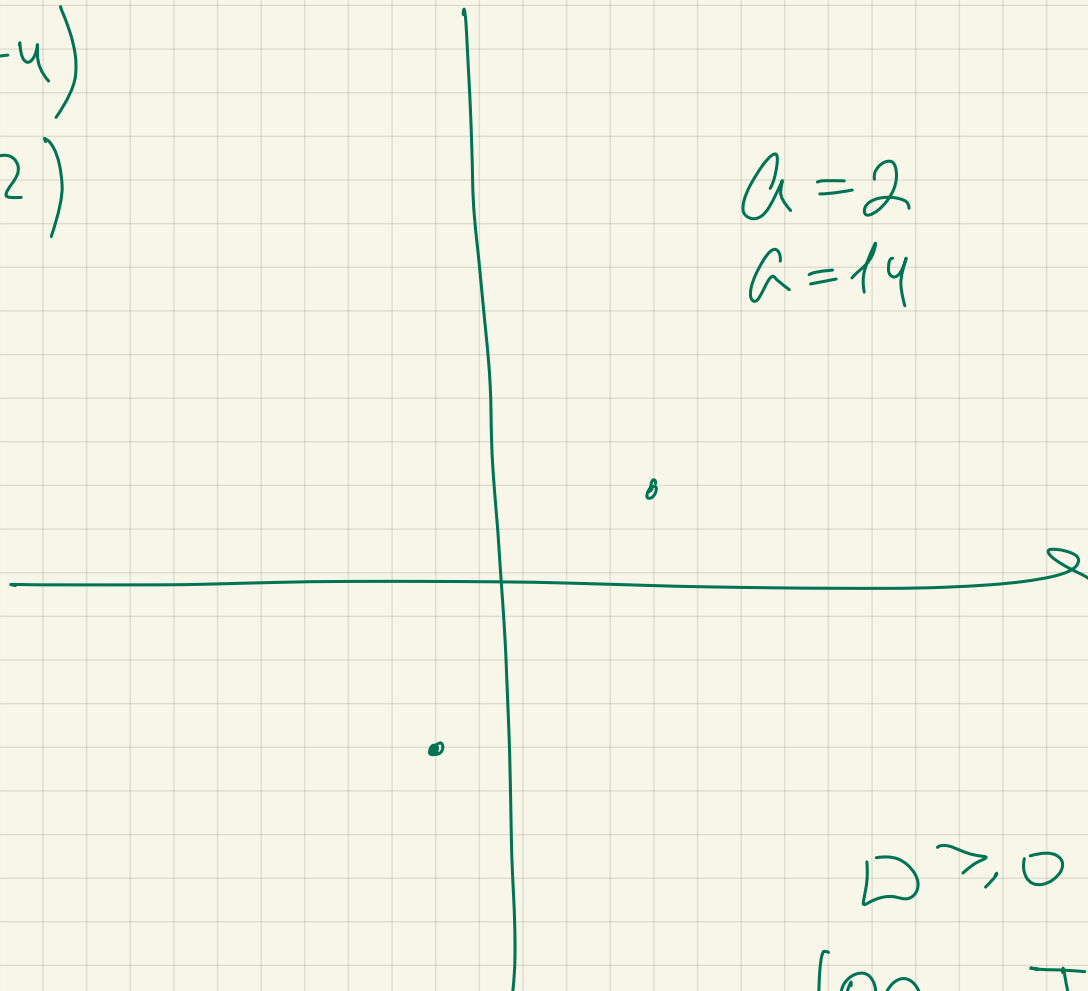
$$a = 8$$

$$a = -8$$

$$\begin{cases} ((x+1)^2 + (y+4)^2) ((x-3)^2 + (y-2)^2) \leq 0 \\ (x-a+1)^2 + (y-2a+2)^2 \leq 4(a-1)^2 \end{cases}$$

me sono per

$$\begin{aligned} &(-1; -4) \\ &(3; 2) \end{aligned}$$



$$\begin{aligned} a &= 2 \\ a &= 14 \end{aligned}$$

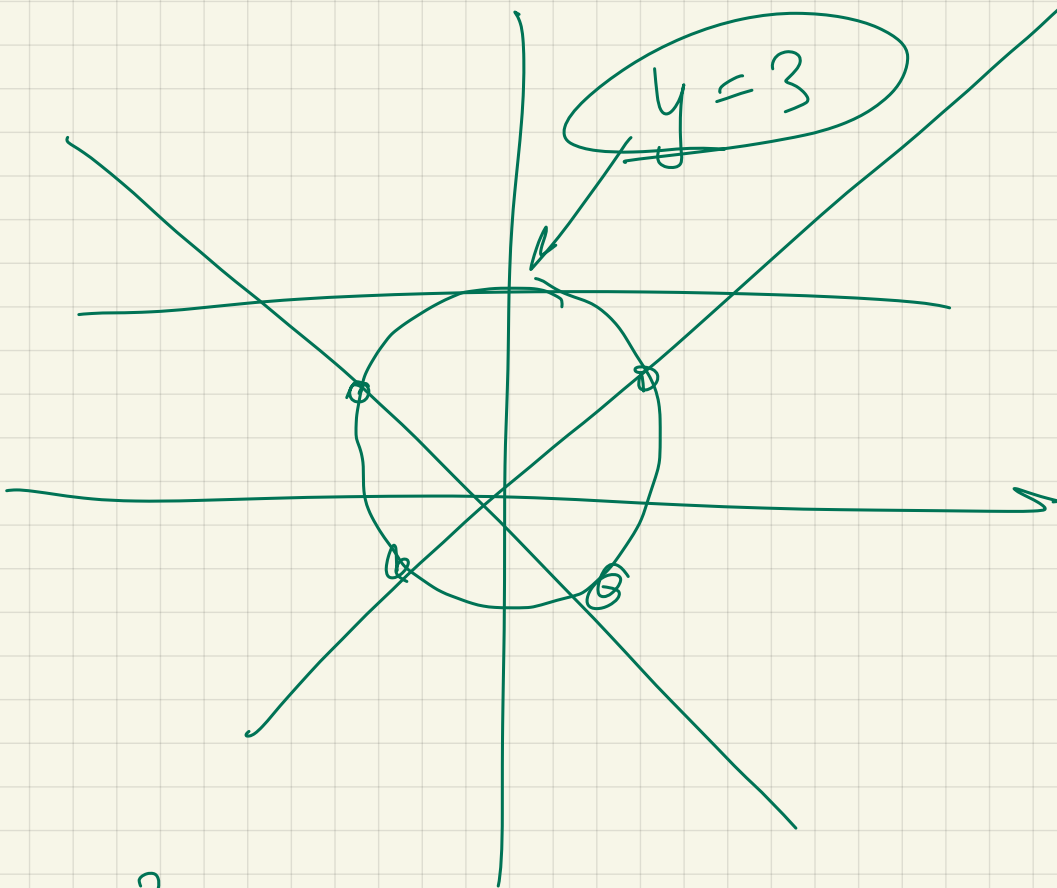
$$D \geq 0$$

$$\underline{\underline{(00, -T) \leq r}}$$

$$a \in (2; 14)$$

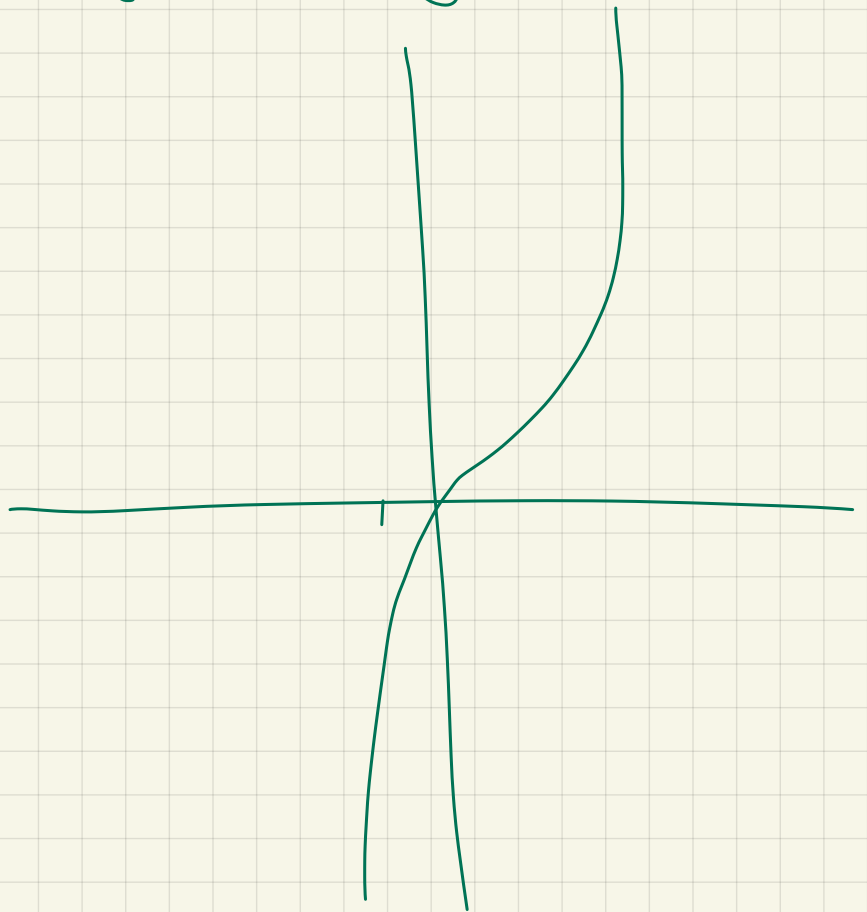
$$a \in (-16; 0)$$

$$a \in (-\infty; -16) \cup (6; 2) \cup (14; +\infty)$$



$$2\sin^2 x - 2\cos 2x - \sin 2x = 0$$

$$\lg x = -\frac{1}{2}$$



$$\lg x = -a$$

$$x = -\operatorname{arclg} a + \pi k$$

$$x = -\operatorname{arclg} \frac{1}{2} + \pi k$$

$$\frac{-\cancel{\frac{1}{4}} + 2 + 3,5 + -10}{\frac{1}{4} + 2 - 2 + +3} \geq 0$$

$$4,31^3 S = 4,0261.69640821$$

$$\frac{210 \ 891393,4}{4,31^3}$$

$$\underline{\underline{S = 93809100}}$$

$$\left\{ \begin{array}{l} (x-1) \left(2x + \frac{2}{5} - 5(1-x) \right) > 0 \\ x > 0 \quad \checkmark \\ x \neq 1 \\ \frac{2x + \frac{2}{5}}{5(1-x)} > 0 \end{array} \right.$$

$$\begin{aligned} & 0,03S + x + 0,03S - 0,03x + 0,03S - 0,03 \cdot 2x \\ & \quad - \dots - 0,03S - 0,03(n-1)x \\ & \quad + 1,03(S - nx) \end{aligned}$$

$$\begin{aligned} & (n-1) \cdot 0,03S + x(n-1) + 1,03(S - nx) \\ & - 0,3x(1+2+\dots+n-1) \end{aligned}$$

$$\left(\frac{1+n-1}{2} \right) n-1 = \frac{n}{2}(n-1)$$

$$\begin{aligned} & 0,03S(n-1) + x(n-1) + 1,03S - 1,03 \cdot nx \\ & - 0,03x \frac{n}{2}(n-1) = \sum \end{aligned}$$

$$\underline{0,03S_n - 0,03S} + \underline{Xn - X} + \underline{1,03S - 1,03nx} \\ - 0,03 \times \frac{n}{2}(n-1) = \Sigma$$

$$S + 0,03Sn - 0,03nx - X - 0,03 \times \frac{n}{2}(n-1)$$

$$X = \frac{S}{n}$$

$$\Sigma = 1,3S$$

$$S + 0,03Sn - 0,03/n \cdot \frac{S}{n} - \frac{S}{n} - 0,03 \frac{S}{n} \cdot \frac{n}{2}(n-1) \\ = 1,3S$$

$$S + 0,03Sn - 0,03S - \frac{S}{n} - 0,03S \frac{(n-1)}{2} = 1,3S$$

$$0,97S + 0,03Sn - \frac{S}{n} - 0,03S \frac{(n-1)}{2} = 1,3S$$

$$0,03n - \frac{1}{n} - 0,03 \frac{(n-1)}{2} = 1,3 - 0,97$$

$$2 \cdot 0,03n^2 - 2 - 0,03n(n-1) = 0,33 \cdot 2 \cdot n$$

$$0,06n^2 - 2 - 0,03n^2 + 0,03n = 0,66n$$

$$0,03n^2 - 0,63n - 2 = 0$$

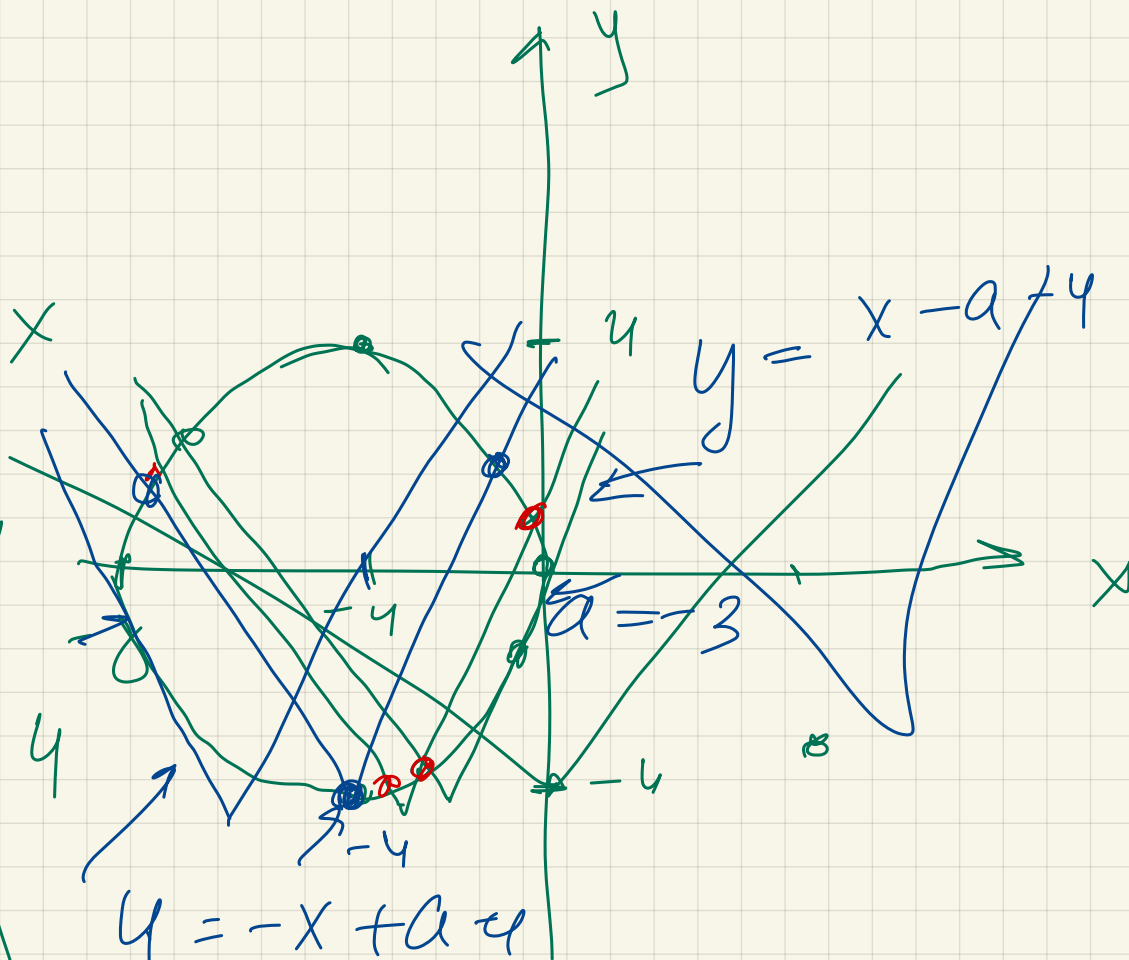
$$y \geq 0$$

$$y = -\frac{x^2}{4} - 2x$$

$$x_0 = \frac{2}{-\frac{2}{4}} = -4$$

$$y_0 = -\frac{16}{4} + 8 = 4$$

$$-x \left(\frac{x}{4} + 2 \right) \quad y = -x + a + 4$$



$$y = |x - a| - 4$$

$$a = 0$$

$$y = |x| - 4$$

$$y \leq 0$$

$$y = \frac{x^2 + 8x}{4} = \frac{x^2}{4} + 2x$$

$$(x - 4) - 4$$

$$x_0 = \frac{-8}{\frac{1}{2}} = -4$$

$$y_0 = \frac{16}{4} - 8 = -4$$

$$\frac{x^2 + 8x}{4} = x - a - 4$$

$$x^2 + 8x = 4x - 4a - 16$$

$$x^2 + 4x + 4a + 16 = 0$$

$$\Delta = 0 \quad 16 - 4(4a + 16) = 0$$

$$4 - 4a - 16 = 0$$

$$-4a - 12 = 0$$

$$\underline{\underline{a = -3}}$$

$$\frac{x^2 + 8x}{4} = -x + a - 4$$

$$a \in (-3; -4) \cup (-4; -5)$$